

Tarun Mittal

Senior Software Engineer

(+91) 9001542418 | tm.eng2021@gmail.com | [linkedin.com/in/t-mittal](https://www.linkedin.com/in/t-mittal) | github.com/TM-DEV-PRO | Bangalore, India

SUMMARY

- Senior Software Engineer with **5 years of experience** designing and owning cloud-native distributed systems.
- Expertise in **Python microservices**, OLAP data platforms (**ClickHouse**, **PostgreSQL**), event-driven systems, and **Agentic AI and RAG**.
- Led backend migrations end to end and **mentored engineers** at Uber, Masters India, and Impact Analytics.

SKILLS

- **Languages:** Python, Go (Golang), SQL.
- **Backend & APIs:** FastAPI, Gin, Django, SQLAlchemy, Pydantic, REST, gRPC, Celery, asyncio.
- **AI & Agents:** LLM agents (OpenAI/Gemini/Claude), LangGraph, LangChain, MCP, RAG, pgvector.
- **Data & Streaming:** Kafka, Flink, Spark, Pinot, ClickHouse, BigQuery, PostgreSQL, MySQL, Redis, MongoDB, Elasticsearch, S3, CDC, CQRS.
- **Cloud & Infra:** GCP, AWS, Docker, Kubernetes, CI/CD, Bazel, Observability (ELK, New Relic, Prometheus, Grafana).
- **Core:** Distributed Systems, Microservices, System Design (HLD/LLD), Concurrency, High Availability, Caching, Mentorship, Unit Testing.

PROFESSIONAL EXPERIENCE

Senior Software Engineer IMPACT ANALYTICS, Bangalore May 2026 – Present

Agentic AssortSmart Platform & PostgreSQL to ClickHouse Data Migration

- Built a **LangGraph/MCP clustering copilot** with 14 audited tools; targets: days to under 1h, 1 to 20+ configurations, 8.5% to under 2% failures.
- Benchmarked a **23.7M-row** Order Batching query at **3.86s on ClickHouse vs 3m 40s+ on PostgreSQL**, a **60x** improvement.
- Sustained **5.9M vs 250K rows/sec** on bulk loads and improved query-to-CSV export throughput **43x**, from 607 to 26.4K rows/sec.
- Designed a **CQRS/CDC migration**: PostgreSQL writes, ClickHouse reads, daily dimension refresh, and **Redis RYW routing with a 30s TTL**.
- Cut 10K-row updates on a **29M-row fact table from 39s to 7s** via partition merges; compressed 17M rows **7.8x** to 433 MiB.
- **Tech Used:** Python, FastAPI, LangGraph, MCP, ClickHouse, PostgreSQL, CDC, Redis, Kafka, GCP, Docker.

Software Development Engineer 2 UBER (via EPAM Systems), Bangalore July 2024 – May 2026

Financial Risk Management (FRM) Scoping Platform (Uber Finance)

- Built Uber's **FRM Risk Scoping platform** across **8 screens and 30+ APIs**, targeting 70% less manual work at **\$340M materiality**.
- Designed an **11-table SQLAlchemy/MySQL schema** using handler, service, and repository layers with header-based authentication.
- Migrated Sheets-backed reconciliation to **MySQL v2 APIs**, validated HFM against 10-Q data, and fixed a column-aliasing bug.
- **Led 3 engineers** and grew the Bazel-tested service to **1,100+ pytest tests** across handlers, services, and repositories.
- **Tech Used:** Python, FastAPI, SQLAlchemy 2.0, Pydantic, MySQL, Bazel, pytest, Docker.

Menu Ingestion & Automation Platform (Uber Eats)

- Built a **Kafka/Flink/Spark** pipeline for 30K+ menus/month; cut onboarding 24h to 2h and saved **\$600K+/year**.
- Built a **RAG/Gemini** extraction pipeline for PDFs and images, reaching **98% fidelity and 100% schema consistency** in offline evaluation.
- Raised ingestion success about 60% to 95%+ via proxies and IP rotation; **Pinot** dashboards cut detection from hours to minutes.
- **Tech Used:** Python, Apache Kafka, Apache Flink, Apache Spark, Apache Pinot, Selenium, Gemini, GCP, Docker.

Software Development Engineer 2

MASTERS INDIA, Noida

December 2022 – June 2024

GST Compliance & E-Invoicing Platform

- Led a PHP-to-**FastAPI microservices** migration and mentored **2 engineers**, cutting p95 latency **1.2s to 300ms** for 1,500+ clients.
- Scaled bulk imports to **100K+ transactions** and 1M+ daily transactions, raising sustained throughput **700 to 4,000 requests/min**.
- Added **Redis cache-aside** to cut redundant DB reads 30%; shipped audit logs that reduced client churn **15%**.
- Added **ELK/New Relic** alerting, cutting triage 70%; raised test coverage **35% to 82%** with a 98% deployment success rate.
- **Tech Used:** Python, FastAPI, PostgreSQL, MongoDB, Redis, Elasticsearch, Celery, Docker, ELK, New Relic, AWS.

Software Development Engineer

GEEKSFORGEEKS, Noida

August 2021 – November 2022

- Migrated PHP services to **Django**, improving reliability for **10K+ daily queries**.
- Built voting and pinning APIs using MySQL, MongoDB, Redis, and Elasticsearch, lifting **premium subscriptions 15 to 20%**.
- Built influencer analytics and cron pipelines for video processing, raising **course sales 30%** and operational efficiency 70%.

EDUCATION

B.Tech (Information Technology), CGPA 7.7/10

Institute of Engineering & Technology,
Lucknow

July 2017 – June 2021

CERTIFICATIONS

- **HackerRank Problem Solving Certificate.**
- **LangChain Academy Certificate.**

ACHIEVEMENTS

- Ranked **2260 / 37,000+** in the Google Code Jam Qualifier Round 2021.
- Finalist (top 3 nationally) at the **Smart India Hackathon 2020 (MHRD)**.
- Participated in the **Global AI Hackathon** at EPAM Systems.